AN EXACT RATIONAL CERTIFICATE FOR ONE-SAMPLE MATROID PROPHET INEQUALITIES

ABSTRACT. This note proves a one-sample matroid prophet guarantee with coefficient $C_{141} = 165971304277649481670619/324956758716416046811841770291200$. The guarantee covers every finite matroid and independent nonnegative values with finite expected offline optimum, even when the arrival-order adversary sees all samples, values, and seed bits. The construction is due to OpenAI, with quantitative refinements due to Ryan Shea. For the unchanged parameters, replacing the conservative estimate $B > 140$ by $B > 141$ gives $C_{141} > C_\star$, where $C_\star$ is Shea's displayed rational certificate. The complete fixed-vector argument, lossless simulation, and exact scalar certificate are included. The result is an exact rational certificate for this construction.

INTRODUCTION AND ATTRIBUTION

OpenAI's manuscript of September 23, 2026 gives a distribution-independent, one-sample matroid prophet rule against an adversary informed of the entire seed [1]. Its abstract states 2^{-310}; the fixed-vector accounting in Section 6.4 derives 2^{-293}, and the one-sample simulation preserves that coefficient. Shea's October 11 refinement [3] gives the explicit certificate

$$C_\star = \frac{408475444978690674223066 1}{800909470202926728724925644800000 0}.$$

The present note organizes the accepted proof for the same model and parameters. Its explicit coefficient is

$$\begin{aligned} C_{141} &= \frac{165971304277649481670619}{324956758716416046811841770291200}, \\ C_{141} - C_\star &= \frac{116935879834753076315538659}{159228811771043862937802467442688000000} > 0. \end{aligned} \tag{1}$$

Only the conservative evaluation changes: the unchanged $B = (261/256)^{256}$ exceeds 141, rather than merely 140. The full attributed formula $C(B)$ is larger still.

The density paths, canonical generator budget, hybrid comparison, layer accounting, fixed-vector model, and one-sample simulation originate in OpenAI [1, Sections 2–6], at commit `1581d4cbdd69a0faf34b2089614fe2a814316a6a`. The quadratic transaction bound, shared phase, projected witnesses, activation charge, three-state encoding, and restricted overlap are Shea's refinements [2, 4], at commit `8cd1395994596d89ad20c2af8b69dab8e3402ee1`. All structural arguments needed here are given below.

Theorem 1. *For every finite labeled matroid $M = (E, \mathcal{I})$, known in advance, there is a measurable distribution-independent online rule with one independent sample per element and exactly $47|E|+27$ initial independent fair bits. Let S_e, V_e be independent nonnegative random variables with the same law D_e at each label, jointly independent across labels and independent of the seed. Assume $\mathbb{E}\,\mathrm{OPT}_M(V) < \infty$. For every measurable arrival-order adversary that sees all samples, all values, and the complete seed, including unused coins, the accepted set is independent after every prefix and*

$$\mathbb{E}[\text{reward}] \geq C_{141}\,\mathbb{E}[\mathrm{OPT}_M(V)].$$

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The assertion includes empty and rank-zero matroids, loops, ties, zeros, and nonrepresentable matroids. Computation and storage may be exponential.

1. Notation, fixed-vector objective, and constants

Fix a total order on the labels of the finite matroid $M = (E, \mathcal{I})$, and put $N = |E|$. Write r for rank, cl for closure, and $L_0 = \mathrm{cl}(\varnothing)$. A plus between sets denotes union, and

$$r(A \mid P) = r(A + P) - r(P), \qquad \mathrm{OPT}_M(w) = \max_{I \in \mathcal{I}} \sum_{e \in I} w_e.$$

The intermediate rule observes a fixed nonnegative finite vector w only on a seed-selected sacrificed set before arrivals, and then on arriving elements outside that set. Its seed R is independent of w. The objective is

$$\mathbb{E}_R \min_{\pi} \mathrm{reward}(w, R, \pi) \geq C(B)\, \mathrm{OPT}_M(w) \geq C_{141}\, \mathrm{OPT}_M(w), \tag{2}$$

where π ranges over all permutations. Keeping this minimum inside the seed expectation permits complete seed visibility.

We recall the finite matroid facts used in the proof. Augmentation gives equal sizes of maximal independent subsets and extension to bases. To obtain rank submodularity, extend a basis of $A \cap B$ to one of A, then to one of $A \cup B$ using elements of $B \setminus A$. The intersection basis together with the last additions is independent in B, so $r(A)+r(B) \geq r(A \cup B)+r(A \cap B)$. Consequently conditional rank decreases as its base increases, closure is increasing and idempotent, and proper inclusion of flats increases rank. Greedy rank-increasing acceptance over a base spans the scanned set and selects its conditional rank. Decreasing-weight greedy has full rank on each weight prefix; summing the successive nonnegative differences of weight levels proves its maximum-weight property. No representation is required.

The algorithm uses

$$\kappa = 164, \quad \rho = \frac{93}{256}, \quad h = \frac{3}{4}, \quad t = \frac{23}{2^{18}}, \quad \theta = \frac{261}{256}, \quad L = 256, \quad B = \theta^L.$$

Analysis constants are

$$\mu = \frac{105}{8}, \quad \nu = \frac{113}{8}, \quad K = \kappa + \nu = \frac{1425}{8}, \quad a = \frac{1}{864}, \quad b = \frac{1}{4000}, \quad v = \frac{1}{624}, \quad d = \frac{253}{250}.$$

Here a, b are denoted u, w in the scalar source. Put

$$p = \rho + (1-\rho)t, \qquad \lambda = -\log p, \qquad \beta = (B-1)/B,$$
$$b_Y = \frac{(2/3)a(1/\rho - 1) + b}{1 + a} + 2v, \qquad \gamma = \rho/4 - b_Y/d, \qquad a_0 = 1/d,$$
$$\varepsilon = \mu\rho/\kappa + a/d, \qquad L_* = \varepsilon + Kt + \frac{Kt + K\rho/\kappa}{B-1}, \quad s = (1-h)(\gamma - L_*/h).$$

The scalar appendix proves

$$\rho[p^{\mu - 1/4} + 3p^{\mu}/v] \leq a, \qquad (1-p)p^{\nu}/v \leq b, \tag{3}$$
$$0 < d < \lambda, \quad 0 < \gamma \leq \rho/4, \quad s > 0, \quad (1-t)(\gamma\kappa + a_0)/K \leq 1/4, \tag{4}$$
$$C(B) := \frac{(1-t)\beta^2 ts}{L(\theta - 1)K} > C_{141}. \tag{5}$$

2. Construction and structural properties

Choose a uniform phase $q \in \{0, \dots, L-1\}$. Round by

$$u_q(x) = \theta^q B^{\lfloor \log_B(x/\theta^q) \rfloor} \quad (x > 0), \qquad u_q(0) = 0.$$

Use rounded weights, with ties broken by the fixed label order. For a fixed phase write $u = u_q(w)$, $W = \mathrm{OPT}_M(u)$, let I^* be its priority greedy optimum, and let m_* be the largest positive rounded nonloop weight, or zero.

Draw independent per-label masks H, D, C, T of rates h, ρ, t, t, a separate fair per-label mask $M_{\max}$, a fair parity, and a branch indicator with probability t of the maximum branch. All choices, including the phase, are independent and made initially. The maximum branch reveals and sacrifices $M_{\max}$, then accepts only the first unmasked positive nonloop above every revealed positive nonloop in priority. If there are at least two positive nonloops, the highest is unmasked and the second highest masked with probability $1/4$; only the highest can then pass, in every order. With one positive nonloop success has probability $1/2$. Expected worst-order rounded reward is therefore at least $m_*/4$.

The main branch reveals and sacrifices $H \cup D \cup C$. For the H-elements of higher priority than e, denoted $H_{>e}$, define the analysis sets

$$U = \{e \notin H : u_e > 0,\ e \notin \mathrm{cl}(H_{>e})\}, \qquad Y = I^* \setminus H.$$

The candidate test uses only H observations and the element's own observed weight. Candidates are nonloops, and $Y \subseteq U$ is independent: a greedy-optimal element is not spanned by any higher-priority elements. Moreover

$$\mathbb{E}_H u(Y) = (1-h)W, \qquad \mathbb{E}_H u(U) = \frac{1-h}{h}\mathbb{E}_H u(\mathrm{greedy}(H)) \le \frac{1-h}{h}W. \tag{6}$$

For the identity on U, condition on all preceding memberships in priority. The span test is fixed, and the candidate and sampled-greedy probabilities are $1-h$ and h times that test. Sum over labels.

Temporarily divide all weights by θ^q. The true groups U_i have weights B^i, $i \in \mathbb{Z}$. List those meeting D as $G_1, \dots, G_m$, in increasing weight, and write $D_j = D \cap G_j$, with analogous notation for C_j, T_j. The listed levels and revealed members are known when needed; the full groups are analysis objects. If $m = 0$, reject everything on the main branch.

For a flat P, let $Q_j(P)$ be the largest flat $Z \supseteq P$ maximizing $f_j(Z) = |D_j \cap Z| - \kappa r(Z)$; first close any nonflat argument. The counting term is supermodular for intersection and closed union, as is negative rank. The closed union of two maximizers is thus a maximizer, giving a largest one. The map is extensive and increasing: for $P \subseteq P'$ and their largest maximizers Q, Q', the intersection is feasible at P and the closed union at P'; optimality and supermodularity force equality, so $Q \subseteq Q'$. Comparison with $\mathrm{cl}(Q+Z)$ gives

$$|D_j \cap (\mathrm{cl}(Q+Z) \setminus Q)| \le \kappa r(Z \mid Q), \qquad Q = Q_j(P). \tag{7}$$

Set $a_j = -3j$, and let $Q_{j,k} = Q_j$ for $k \ge a_j$, the identity otherwise. Start $F_0(k) = L_0$ for $k < 0$, and $F_0(k) = E$ for $k \ge 0$. Recursively set

$$X_j(k) = Q_{j,k}(F_{j-1}(k)), \qquad F_j(k) = \mathrm{cl}\big(X_j(k) + (C_j \cap X_j(k+1))\big).$$

The matroid and revealed observations determine all these flats before arrivals. The increasing extensive maps imply

$$F_{j-1}(k) \subseteq X_j(k) \subseteq F_j(k) \subseteq X_j(k+1). \tag{8}$$

All paths equal E for $k \ge 0$. Induction gives $X_j(k) = L_0$ for $k < a_j$ and $F_j(k) = L_0$ for $k < a_j - 1$. Since $a_{j-1} = a_j + 3$, the incoming path is L_0 for $k < a_j + 2$, also directly for $j = 1$. Thus

$X_j(a_j) = X_j(a_j+1) = Q_j(L_0)$. Every nonloop birth $b_j(e) = \min\{k : e \in X_j(k)\}$ is a_j or lies in $[a_j+2, 0]$.

There is a canonical set $J_j \subseteq D_j$, with increasing subsets $J_j(k)$, such that

$$X_j(k) = \mathrm{cl}(F_{j-1}(k) + J_j(k)), \qquad |J_j| \le |D_j|/\kappa. \tag{9}$$

Indeed, process $k = a_j, \dots, 0$ and put $P_k = \mathrm{cl}(F_{j-1}(k) + X_j(k-1))$. Optimality gives

$$\kappa[r(X_j(k)) - r(P_k)] \le |D_j \cap X_j(k)| - |D_j \cap P_k| \le |D_j \cap X_j(k)| - |D_j \cap X_j(k-1)|.$$

Also $\mathrm{cl}(F_{j-1}(k) + (D_j \cap X_j(k))) = X_j(k)$: a proper subflat would retain the same D_j count with smaller rank and improve the objective. Choose a fixed-label greedy basis extension from P_k to $X_j(k)$ within $D_j \cap X_j(k)$, and add it to J_j. Earlier generators together with $F_{j-1}(k)$ span P_k, proving the representation. The displayed counts telescope to the budget. Each generator has one insertion time $\tau(g)$. Activation generators $J_{j,0}$, inserted at a_j, form a basis of $Q_j(L_0)$, so $j_0 := |J_{j,0}| = r(Q_j(L_0))$. No generator is inserted at $a_j + 1$.

Put $\widehat{F} = F_m$. For endpoints b of the chosen parity, maintain A_b independent over $\widehat{F}(b-2)$. Reject sacrificed elements, noncandidates, and unlisted weights. An arriving element in listed group j, with birth $b = b_j(e)$, is eligible only if $e \in T$, $b \ge a_j + 2$, b has the chosen parity, and $e \notin \widehat{F}(b-2)$. Accept it exactly when it increases rank over $\widehat{F}(b-2) + A_b$.

Every eligible element in layer $(b-2, b]$ lies in $X_j(b) \subseteq \widehat{F}(b)$; earlier layers lie in each later layer's base. Independence over these nested bases proves that the union of the A_b is independent at every prefix, for every interleaving. Finite flat enumeration, finitely many listed levels and times, fixed ties, and countably many rounding intervals ensure measurable decisions. No future weight is used, zeros are excluded before logarithms, and loops fail the candidate test.

Masks $H, D, C, T, M_{\max}$ use respectively $2, 8, 18, 18, 1$ fair bits per label. Phase, branch, and parity use $8, 18, 1$ shared bits, totaling $47N + 27$. The sacrificed mask depends only on the seed; unused coins can also be disclosed.

3. Reverse exposure and safety

Fix the rounded vector, H, D, and a listed prefix $1, \dots, j$. For $k < 0$, a state $S = (S_\ell)_{\ell \le j}$, and bits c, define

$$A_0 = L_0, \qquad N_\ell = Q_{\ell,k}(A_{\ell-1}), \qquad A_\ell = \mathrm{cl}(N_\ell + \{e \in G_\ell \cap S_\ell : c_e = 1\}).$$

Let $\mathcal{G}_k(c, S) = (N_\ell)_{\ell \le j}$. It is increasing in time, state, and bits. Starting from $X_\ell(0) = E$, the recursion is $X(k) = \mathcal{G}_k(C, X(k+1))$. For an actual later state $S = X(k+1)$, every bit assignment gives $N_\ell \subseteq S_\ell$: inductively

$$A_{\ell-1} \subseteq S_{\ell-1} \subseteq F_{\ell-1}(k+1), \qquad N_\ell \subseteq Q_{\ell,k+1}(F_{\ell-1}(k+1)) = S_\ell,$$

and all additions lie in S_ℓ.

After computing N_ℓ, query only bits on $G_\ell \cap (S_\ell \setminus N_\ell)$, in fixed label order. Each batch is determined by earlier answers, and additions inside N_ℓ do not change closure. These queries determine the output. The domains shrink, so no bit is queried twice. A predictable exposure of product bits leaves unqueried bits independent with their original laws: induction on the transcript applies because the next coordinate is determined by known data and its answer specifies only that coordinate.

Let $\mathcal{H}_b$ be the transcript immediately before $X(b) \to X(b-1)$. The remaining bits are exactly on $V_b = \bigcup_{\ell \le j}(G_\ell \cap X_\ell(b))$, conditionally independent Bernoulli(t). For a nonloop e alive in $X_j(b)$, put

$$p_{b-1} = \mathbb{P}(e \notin X_j(b-1) \mid \mathcal{H}_b),$$

and put it zero when e is dead. Every nonloop exits by $X_j(a_j - 1) = L_0$.

Lemma 2 (Quadratic transaction bound)**.** *Let A be an increasing event of finitely many independent bits, decided by a predictable procedure reading each bit at most once. If $p = \mathbb{P}(A^c)$, and q is the terminal failure probability when queried bits are reset to zero and the unqueried bits retain their laws, then*

$$\mathbb{E}[\mathbf{1}_A q^2] \geq p(1-p). \tag{10}$$

Proof. At each transcript let m be the actual conditional failure probability and q its reset version. Monotonicity gives $0 \leq m \leq q \leq 1$. Initially $m = q = p$. On the next query, q' is known before the answer and $q' \geq q$, while $\mathbb{E}[m' \mid \text{transcript}] = m$. Thus for $\Phi(q, m) = q^2 - 2qm$,

$$\mathbb{E}[\Phi(q', m') \mid \text{transcript}] - \Phi(q, m) = (q' - q)(q' + q - 2m) \geq 0.$$

At completion $m = \mathbf{1}_{A^c}$, and on failure $q = 1$. The initial potential is $-p^2$ and the terminal expected potential is $\mathbb{E}[\mathbf{1}_A q^2] - p$. Sum over the bounded number of queries, padding stopped paths with zero increments. This also covers $p = 0, 1$ and no queries. □

Apply the lemma conditionally on $\mathcal{H}_b$ and being alive, to survival of the reverse transition. If S' is its output and V' the remaining domains, the next hazard r, on survival, is at least the depleted probability q. Indeed, for every c supported on V',

$$\mathcal{G}_{b-2}(c, S') \subseteq \mathcal{G}_{b-1}(c, S).$$

The right side resets precisely the just-queried coordinates to zero; the left uses the next transition with its frozen actual input, without recomputing earlier states. Remaining bits retain the product law. Therefore

$$\mathbb{E}[\mathbf{1}_{\text{survive}} r^2 \mid \mathcal{H}_b, \text{alive}] - p_{b-1}^2 \geq p_{b-1} - 2p_{b-1}^2.$$

Telescope the alive squared-hazard potential over the finite horizon, ending at zero. The expected sum of alive hazards is one, because there is exactly one exit. If P is the hazard at exit, conditioning at each step shows that $\mathbb{E}P$ is the expected sum of alive squared hazards. Consequently

$$-p_{\text{start}}^2 \geq 1 - 2\mathbb{E}P, \qquad \sum_b \mathbb{E}[\mathbf{1}_{\{e \in X_j(b)\}} p_{b-1}^2] = \mathbb{E}P \geq \frac{1}{2}. \tag{11}$$

For a valid endpoint $b \geq a_j + 2$, define lower test competitors

$$K_{j,b} = \bigcup_{\ell < j} [T_\ell \cap (X_\ell(b) \setminus X_\ell(b-1))] \subseteq F_{j-1}(b).$$

At a reverse step $S = X(b) \to S' = X(b-1)$, form hybrid bits using C on S'_ℓ and T on $S_\ell \setminus S'_\ell$, within each G_ℓ. Conditional on the complete ordinary-step transcript, retained C bits have never been queried, and T is independent of guards. The hybrid input is therefore a fresh product Bernoulli(t) vector at the same frozen input S. In particular,

$$\mathbb{P}(e \text{ alive, ordinary exit, hybrid exit} \mid \mathcal{H}_b) = \mathbf{1}_{\{e \text{ alive}\}} p_{b-1}^2. \tag{12}$$

For the hybrid nominal output $\widetilde{N}$, deterministically

$$Q_j(F_{j-1}(b-2) + K_{j,b}) \subseteq \widetilde{N}_j. \tag{13}$$

To see this, induct over $\ell < j$. Time and state monotonicity put $X_\ell(b-2)$ in its hybrid nominal flat. Retained guards include $C_\ell \cap X_\ell(b-1)$, so the intermediate flat contains $F_\ell(b-2)$. Departing test additions include every T-positive in $G_\ell \cap (X_\ell(b) \setminus X_\ell(b-1))$, and extensivity preserves earlier additions. The final incoming intermediate flat contains $F_{j-1}(b-2) + K_{j,b}$; the focal density map is enabled. This is pointwise and does not condition on unexposed earlier flats or require independence between hybrid experiments at different endpoints.

Let σ_j count labels in $D_j \cap Y$ whose birth is valid, has chosen parity, and lies outside $Q_j(F_{j-1}(b-2) + K_{j,b})$. Ordinary exit at b means birth b; double exit implies this exclusion by (13). For $e \in D_j \cap Y$ outside $Q_j(L_0)$, every birth is valid. Equations (11)–(12) and fair parity give safety

probability at least 1/4. At most $j_0 = r(Q_j(L_0))$ labels of the independent Y lie in the activation flat. Hence

$$\mathbb{E}[\sigma_j \mid H, D] \geq \frac{|D_j \cap Y| - j_0}{4}. \tag{14}$$

Set σ, j_0 to zero on unlisted true groups. For a true group U_i, with $y = |Y \cap U_i|$, averaging D gives

$$\mathbb{E}[\sigma \mid H] \geq \rho y/4 - \mathbb{E}[j_0 \mid H]/4. \tag{15}$$

4. Projected residuals and marked streams

Fix the phase and H. For a listed group j, set

$$P_{j,b} = G_j \cap (X_j(b) \setminus X_j(b-1))$$

at valid chosen-parity endpoints $a_j + 2 \leq b \leq 0$, and take it empty at other endpoints of that parity. Write $O = D \cup C$, and define

$$z_j^{\text{full}} = \sum_b r(P_{j,b} \setminus O \mid X_j(b-2) + K_{j,b}).$$

For analysis, choose a fixed-label basis extension $Z_b \subseteq (P_{j,b} \cap Y) \setminus O$ over $X_j(b-2) + K_{j,b}$. Put $Z = \bigcup_b Z_b$ and $z = \sum_b |Z_b|$. Disjoint births give $|Z| = z \leq z_j^{\text{full}}$. These witnesses are not algorithmic inputs and use no focal T_j bit. The projected residual

$$R_Y = \bigcup_b [(P_{j,b} \cap Y) \setminus \mathrm{cl}(X_j(b-2) + K_{j,b} + Z_b)]$$

lies in O by the witness spanning property. For a safe sample at b outside R_Y, put $Q'_b = Q_j(F_{j-1}(b-2) + K_{j,b})$. Monotonicity gives $X_j(b-2) + K_{j,b} \subseteq Q'_b$. The sample is in $D_j \cap (\mathrm{cl}(Q'_b + Z_b) \setminus Q'_b)$, so (7) bounds such samples by $\kappa|Z_b|$. Thus

$$\sigma \leq \kappa z + |R_Y|. \tag{16}$$

We construct a deterministic family containing all residuals before exposing focal mask bits. In addition to H and parity, fix all D, C, T bits outside a true group U of size n. Put $Y_U = Y \cap U$, $y = |Y_U|$. If listed, its index j is one plus the number of listed lower groups; this potential index is fixed without conditioning on listing. The incoming $F(k) = F_{j-1}(k)$, activation a_j, and $K_b = K_{j,b}$ are fixed. The generator representation reads

$$X(k) = \mathrm{cl}\big(F(k) + \{g \in J : \tau(g) \leq k\}\big).$$

Split $J = J_0 \cup J_1$ into activation and active generators, of sizes j_0, j_1. Since $J \subseteq D \cap U$ and $Z \subseteq Y_U \setminus O$, these role sets are disjoint.

Consider streams of labeled occurrences, with marks zero or one; repeated labels are allowed. The pivot of a label is the first occurrence after whose insertion it is spanned, and its pivot mark is that occurrence's mark. The first stream runs through $k = a_j - 2, \ldots, 1$. At time k insert $F(k)$ in label order, marked one exactly at valid chosen-parity endpoints, followed by the generators with $\tau(g) = k$ in label order and with the same mark. Its span after time k is $X(k)$. The mark-one set on U is precisely the union of valid birth sets P_b.

For the second stream, let b_0 be the chosen-parity member of $\{a_j - 2, a_j - 1\}$, and b_1 that of $\{0, 1\}$. Start with $F(b_0)$, marked zero. At successive endpoints $b = b_0 + 2, \ldots, b_1$, insert first K_b, then Z_b, both in label order with mark zero; then insert $F(b)$ in label order and all generators with $b - 2 < \tau(g) \leq b$, with mark one. Put $K_b = Z_b = \varnothing$ at invalid endpoints. The intermediate and endpoint spans are respectively

$$\mathrm{cl}(X(b-2) + K_b + Z_b), \qquad X(b),$$

because $K_b \subseteq F(b)$ and $Z_b \subseteq X(b)$. An element passing the first test with birth b is outside $X(b-1)$, hence outside $X(b-2)$. Its second pivot occurs in this same layer and has mark one

exactly when it is outside the intermediate span. The intersection of the two mark-one sets on Y_U is therefore R_Y. Reordering a second-stream generator block preserves this property, as all its marks are one and its endpoint span is fixed.

5. Reconstruction and three-state encoding

The following reconstruction and joint encoding implement the refinements of [2, 4].

Fix an ordered list of old occurrences and insert specified movable occurrences in arbitrary positions and order. If the greedy scan retains exactly a set T of movable occurrences, the retained old occurrences are exactly the greedy scan of the old list after contracting their labels. Indeed, induct over old occurrences, with A the earlier retained old ones. A rejected old occurrence is spanned by A and previously retained movable occurrences, hence by $A + T$. If retained, the final independent set contains $A + T + \{e\}$, so it increases rank over $A + T$. These are exactly the contracted scan decisions. Occurrences remain distinguished, and repeated labels or loops cannot both be retained. Thus fixing T determines the retained old occurrences, also for lists with repetitions.

Every canonical generator is retained in the first stream: immediately before its insertion, the span is $\mathrm{cl}(F(k) + X(k-1))$ plus its batch predecessors, over which it was selected to increase rank. Every witness is retained in the second stream because Z_b is independent over $X(b-2) + K_b$. Reordering the following generator block does not affect this. Activation generators have fixed positions: negative mark at a_j in the first stream and positive mark at endpoint a_j or $a_j + 1$ in the second. There the incoming flat is L_0 and competitors and witnesses are empty, so they are retained in both streams. Include them among the fixed old occurrences.

Partition the remaining generators into

$$J_+ = \{\text{retained in the second basis, inserted at valid chosen parity}\},$$
$$J_- = \{\text{retained in the second basis, inserted at the other parity}\},$$
$$J_o = \{\text{omitted from the second basis}\}.$$

Fix disjoint roles J_0, J_+, J_-, J_o, Z. The first retained old set is obtained by contracting $J_1 = J_+ \cup J_- \cup J_o$; the second by contracting $J_+ \cup J_- \cup Z$. Both retained bases of occurrences are fixed, with no further basis-choice factor. Impossible assignments can be discarded; including extras only overcounts.

Here is an explicit joint affine encoding using $j_1 + z$ variables, independent of timeline length. Label E by $0, \dots, N-1$, put $m = N + 1$, $Q = 8m$, and $U_k = (k - a_j + 2)Q$. In the first stream the old $F(k)$ occurrence of label e has absolute score 2^{U_k+4m+e}, positive at valid chosen-parity endpoints and negative otherwise. In the second, assign negative scores $-2^{U_{b_0}+4m+e}$ to initial $F(b_0)$, negative scores -2^{U_b+e} to K_b, and positive scores 2^{U_b+4m+e} to $F(b)$. Activation generator scores are fixed: $-2^{U_{a_j}+6m+g}$ in the first, and 2^{U_c+6m+g} in the second, where $c \in \{a_j, a_j + 1\}$ has chosen parity.

For an active generator let $x_g = 2^{U_{\tau(g)}+6m+g}$. If $g \in J_+$, it contributes x_g in both streams. If $g \in J_-$, its scores are $-x_g$ in the first and $2^Q x_g$ in the second, shifting insertion to the next chosen-parity endpoint. If $g \in J_o$, use one signed variable equal to its actual signed first score; it contributes no term in the retained second basis. For $e \in Z_b$, use $y_e = 2^{U_b+2m+e}$ and score $-y_e$ only in the second stream. There are $j_1 + z$ common real variables. Active insertions are at least $a_j + 2$, so there is no extra invalid-parity type.

Order the second-stream generator block by label. Its shifted scores are then 2^{U_b+6m+g}, consistent with numeric order; define states using the retained bases of this reordered stream. In each stream all actual absolute scores are distinct powers of two in chronological order, each exceeding the sum of earlier absolute scores. The $2m$ block gaps accommodate all labels, and $Q = 8m$ separates times. The reordering preserves pivot marks and witness retention.

For any spanning basis B_* of occurrences and nonloop f, there is a unique minimal subset $S_f \subseteq B_*$ spanning f. If S, T span f, independence and rank submodularity applied to $S+\{f\}, T+\{f\}$ show that $S \cap T$ spans f; intersect all spanning subsets. The support is nonempty. A retained greedy prefix spans the full prefix, so the pivot of f is the last retained member of S_f. The sign of the sum of support scores is its pivot mark, since the last score dominates all earlier ones.

For fixed roles and states, the two supports for each $f \in Y_U$ are fixed. Their sums are $2y$ affine functions of the same $j_1 + z$ variables, with old and activation scores constant. Every actual mark vector is a strict sign pattern. Allowing arbitrary real variable values, including dropping positivity restrictions, enlarges the family.

For M affine functions in $\mathbb{R}^d$, the number of strict sign patterns is at most

$$S_d(M) = \sum_{i=0}^{\min(d,M)} \binom{M}{i}.$$

Each strict pattern is an open convex region. A new proper hyperplane splits at most as many regions as the arrangement induced on that hyperplane. Thus $N(M,d) \le N(M-1,d) + N(M-1,d-1)$, with $N(0,d) = 1$, $N(M,0) \le 1$. Nonzero constants have fixed sign, identically zero functions yield no all-strict pattern, and coincidences or degeneracies cannot increase the bound. Induction gives the sum. Hence each fixed role assignment yields at most

$$S_{j_1+z}(2y) \tag{17}$$

residuals. A generator retained in both bases has itself as pivot in each. Thus J_+ elements in Y_U are residual and J_- elements are not. Activation elements have negative first pivots; witnesses have negative second pivots. For actual residuals,

$$R_Y \cap J \subseteq (J_+ \cup J_o) \cap Y_U, \qquad R_Y \cap (J_0 \cup J_- \cup Z) = \varnothing. \tag{18}$$

Omitted generators can be residual and remain in the overlap correction. All roles, states, old lists, coefficients, and supports specifying this family are fixed before focal masks are exposed. The family may contain unrealizable streams. No independence of adaptively chosen generators or witnesses from those masks is assumed.

6. Joint mask law and residual tail

Keep H, parity, and masks outside U fixed. For disjoint roles J_0, J_+, J_-, J_o, Z, restrict the fixed family to $R \subseteq Y_U \setminus Z$ satisfying (18). Independent focal pairs of D, C bits give exactly

$$\begin{aligned}&\mathbb{P}(J \subseteq D,\ Z \cap O = \varnothing,\ R \subseteq O)\\ &\quad = \rho^{|J|}(1-p)^z p^{|R\setminus J|} \le \rho^{|J|}(1-p)^z p^{|R|-|(J_+\cup J_o)\cap Y_U|}.\end{aligned} \tag{19}$$

Labels in J are already in $D \subseteq O$; those in Z must be outside O; the other labels of R must be inside O. These are disjoint coordinate sets. Each allowed overlap costs $1/p$, including omitted generators. This is a law for fixed witnesses, not an independence assertion for the selected residual.

On listed outcomes put

$$e_{\mathrm{err}} = \big[|R_Y| - (\mu - 1/4)j_0 - \mu j_1 - \nu z\big]_+,$$

and set it zero on unlisted outcomes. If $e_{\mathrm{err}} > x$, some fixed-role, fixed-family witness has $|R|$ exceeding its charge plus x. Apply (19) and union bound. For an integer dimension D',

$$v^{D'} S_{D'}(2y) \le \sum_{i=0}^{\min(D',2y)} \binom{2y}{i} v^i \le (1+v)^{2y},$$

since $0 < v < 1$. Thus (17) costs $1/v$ for each active generator or witness. Put

$$\alpha_0 = \rho p^{\mu - 1/4}, \qquad \alpha_1 = \rho p^{\mu}/v, \qquad \omega = (1-p)p^{\nu}/v.$$

A label outside Y_U has alternatives no role, activation, or three active states, of total weight $1+\alpha_0+3\alpha_1$. Inside Y_U, the witness role is also possible, and the two potentially overlapping states J_+, J_o receive $1/p$. Multiplying the per-label totals sums all disjoint role assignments and gives

$$\mathbb{P}(e_{\mathrm{err}} > x) \le \min\{1, Mp^x\},$$
$$M \le (1+\alpha_0+3\alpha_1)^{n-y}(1+\alpha_0+\alpha_1+2\alpha_1/p+\omega)^y(1+v)^{2y}. \tag{20}$$

All sizes and placements are included, with no extra timeline, subset, or order factor. Dropping generator budgets only overcounts.

The tilts (3) imply $\alpha_0+3\alpha_1 \le a$, $\alpha_1 \le a/3$, and $\omega \le b$. Replace M by

$$M_* = (1+a)^{n-y}\big[1+a+(2/3)a(1/p-1)+b\big]^y(1+v)^{2y} \ge 1.$$

Integration of $\min\{1, M_*\exp(-\lambda x)\}$ over $[0,\infty)$ gives $(\log M_*+1)/\lambda$. Since $p \ge \rho$ and $\log(1+x) \le x$ for $x \ge 0$,

$$\begin{aligned}\log M_* &= n\log(1+a) + y\log\left(1+\frac{(2/3)a(1/p-1)+b}{1+a}\right) + 2y\log(1+v)\\ &\le na + yb_Y.\end{aligned}$$

Using $d \le \lambda$, then averaging the outside conditioning,

$$\mathbb{E}[e_{\mathrm{err}} \mid H] \le (na + yb_Y + 1)/d. \tag{21}$$

Pointwise, (16) and the definition of the error give

$$\sigma + j_0/4 \le (\kappa+\nu)z + \mu(j_0+j_1) + e_{\mathrm{err}}.$$

On unlisted outcomes all statistics vanish. Also $j_0 + j_1 \le |D \cap U|/\kappa$ always. Combine (15), (21), and $\mathbb{E}|D \cap U| = \rho n$ to obtain

$$K\mathbb{E}[z \mid H] \ge (\rho/4 - b_Y/d)y - (\mu\rho/\kappa + a/d)n - 1/d.$$

Since projected rank is at most full nominal rank, for true group size $n_i = |U_i|$,

$$\mathbb{E}[z_i^{\mathrm{full}} \mid H] \ge \frac{\gamma|Y \cap U_i| - \varepsilon n_i - a_0}{K}. \tag{22}$$

In particular this holds when $n_i \ge \kappa$. All averaging precedes conditioning on listing. The safety estimate averaged guards and tests given H, D; the tail estimate fixed outside masks and then averaged. Only expectations conditional on H are combined.

7. Order-uniform rank and one path-expansion budget

For a listed group j, put $S_{j,b} = P_{j,b} \setminus O$ and

$$\lambda_j = \sum_b r(T_j \cap S_{j,b} \mid \widehat{F}(b-2) + K_{j,b}).$$

This is independent of arrival order. Let $N_{\ge j}(\pi)$ count accepted elements of rounded weight at least that of G_j. Simultaneously for every seed and every permutation,

$$N_{\ge j}(\pi) \ge \lambda_j. \tag{23}$$

Indeed, in a layer with base $F = \widehat{F}(b-2)$, the final accepted set B_b is a basis of the full eligible set E_b over F, regardless of interleaving with other layers. Partition it into $L_b + H_b$ with weights below G_j, and the rest. Independence and spanning imply $|H_b| = r(E_b \mid F + L_b)$. Every label of $T_j \cap S_{j,b}$ outside F is eligible; labels inside F add no rank. Also $L_b \subseteq K_{j,b}$, because lower accepted labels are T-positive and have birth b in their lower groups. Thus

$$|H_b| \ge r(T_j \cap S_{j,b} \mid F + L_b) \ge r(T_j \cap S_{j,b} \mid F + K_{j,b}).$$

Summing proves (23) for all orders.

To replace nominal by final paths, use the single generator set

$$J^{\mathrm{later}} = C_j \cup \bigcup_{\ell>j}(C_\ell \cup J_\ell), \qquad c_j = |C_j| + \sum_{\ell>j}(|C_\ell| + |D_\ell|/\kappa).$$

Its size is at most c_j, and induction through later groups using (9) and their guards gives, for every integer k,

$$X_j(k) \subseteq \widehat{F}(k) \subseteq \mathrm{cl}(X_j(k) + J^{\mathrm{later}}). \tag{24}$$

For a fixed b, put $A = X_j(b-2) + K_{j,b}$, $S = S_{j,b}$, $J = J^{\mathrm{later}}$. The loss from the final base is at most

$$\begin{aligned} r(S \mid A) - r(S \mid A + J) &= r(J \mid A) - r(J \mid A + S) \\ &\le r(J \mid X_j(b-2)) - r(J \mid X_j(b)). \end{aligned}$$

Here $X_j(b-2) \subseteq A \subseteq A + S \subseteq X_j(b)$, since $K_{j,b} \subseteq F_{j-1}(b) \subseteq X_j(b)$. Sum over valid endpoints of one parity. The differences are nonnegative, and adding missing endpoints permits telescoping, at total cost at most $r(J) \le |J| \le c_j$. Hence

$$\sum_b r(S_{j,b} \mid \widehat{F}(b-2) + K_{j,b}) \ge z_j^{\mathrm{full}} - c_j. \tag{25}$$

There is no independence assumption on J, and no repeated per-layer charge.

Now condition on H, D, C, parity, and test bits outside G_j. Paths, S, competitors, and c_j are fixed, while T_j remains independent Bernoulli(t). For each unthinned rank choose a fixed independent witness over its base. Its intersection with T_j stays independent with expected size t times that rank. Thus

$$\mathbb{E}[\lambda_j \mid \text{this conditioning}] \ge t(z_j^{\mathrm{full}} - c_j). \tag{26}$$

Neither the nominal statistic nor projected witnesses use focal test bits. Extend $c_i, \lambda_i, z_i^{\mathrm{full}}$ by zero on unlisted true groups. Bound costs pointwise by counts from all true groups before averaging masks, obtaining

$$\mathbb{E}[c_i \mid H] \le t n_i + (t + \rho/\kappa) \sum_{\ell>i} n_\ell.$$

Equations (22) and (26) yield, for $n_i \ge \kappa$,

$$\mathbb{E}[\lambda_i \mid H] \ge \frac{t}{K}\left[\gamma|Y \cap U_i| - (\varepsilon + Kt)n_i - (Kt + K\rho/\kappa)\sum_{\ell>i} n_\ell - a_0\right]. \tag{27}$$

8. Weighted payoff, maximum branch, and phase

Use normalized weights B^i at fixed phase, and let $\mathcal{A}(H) = \{i : n_i \ge \kappa\}$. Every accepted weight B^ℓ contributes at most $\sum_{i\le\ell} B^i = B^\ell/\beta$ to weighted cumulative counts. By (23), before any mask expectation,

$$\min_\pi u(A_{\mathrm{main}}(R,\pi)) \ge \beta \sum_{i\in\mathcal{A}(H)} B^i \lambda_i. \tag{28}$$

For every H, geometric sums give

$$\begin{aligned} &\sum_{i\in\mathcal{A}(H)} B^i \sum_{\ell>i} n_\ell \le \frac{u(U)}{B-1}, \\ &\sum_{i\notin\mathcal{A}(H)} B^i |Y \cap U_i| \le \frac{\kappa m_*}{\beta}, \qquad \sum_{i\in\mathcal{A}(H)} B^i \le \frac{m_*}{\beta}. \end{aligned}$$

For the first, exchange the finite nonnegative group sums and bound the lower-level sum by $B^{\ell}/(B-1)$. For the others, occupied nonloop levels are at most m_*, excluded groups have fewer than κ labels, and the geometric sum over all integer levels up to m_* is m_*/β.

Multiply (27) by B^i, sum over $\mathcal{A}(H)$, apply (28), and average H using (6). All error coefficients are nonnegative. The result is

$$\mathbb{E}_R \min_{\pi} u(A_{\mathrm{main}}(R,\pi)) \geq \frac{\beta t s}{K} W - \frac{t(\gamma\kappa + a_0)}{K} m_*. \tag{29}$$

The expectation here includes main masks and parity at fixed phase. Restoring θ^q preserves it. Maximum-branch expected worst-order reward is at least $m_*/4$. Since order can depend on the branch and unused coins, minimize for each full seed, then average the branches. With (4),

$$\begin{aligned}\mathbb{E}_R \min_{\pi} u(A(R,\pi)) &\geq \frac{(1-t)\beta t s}{K} W + \left[\frac{t}{4} - \frac{(1-t)t(\gamma\kappa + a_0)}{K}\right] m_* \\ &\geq \frac{(1-t)\beta t s}{K} W. \end{aligned}\tag{30}$$

For $x > 0$, let $k = \lfloor \log_\theta x \rfloor$. Over all phases the rounded exponents are $k, k-1, \ldots, k-L+1$, once each. As $\theta^k \geq x/\theta$, their geometric sum gives

$$\mathbb{E}_q u_q(x) \geq \frac{\beta x}{L(\theta - 1)}.$$

Choose one true optimal independent set of w and apply this elementwise. The expected rounded optimum W is at least the same factor times $\mathrm{OPT}_M(w)$. Actual accepted weights dominate rounded ones. Averaging (30) over phase proves (2) with $C(B)$ in (5).

Every step used only finite matroid rank and closure. Flat enumeration, rank tables, and storage may be exponential. Exact observations and countably many rounding intervals make the rule measurable; occupied integer levels may be unbounded in index. If there are no positive nonloops the optimum is zero, including empty and rank-zero cases. Loops are rejected, ties follow the label order, and zeros never enter logarithms. Prefix feasibility holds for every seed and permutation, with the complete $47N + 27$-bit seed already specified.

9. Lossless one-sample simulation

Use this fixed-vector seed independently of samples and values. Let $B_{\mathrm{obs}}(R)$ be the sacrificed mask, $M_{\max}$ on the maximum branch and $H \cup D \cup C$ on the main branch. Glue the vector

$$\widetilde{W}_e = \begin{cases} S_e, & e \in B_{\mathrm{obs}}(R), \\ V_e, & e \notin B_{\mathrm{obs}}(R). \end{cases}$$

Conditional on the complete seed, each choice of copy is fixed, so the law of $\widetilde{W}$ is $\prod_e D_e$, independent of the seed. Thus $\widetilde{W}$ is independent of R and has the law of V. Reveal samples only on B_{obs}, reject those labels, and ignore their arriving values. At other arrivals use V_e and ignore the unused sample. This uses one available sample per label and no distributional information.

For each fixed S, V, R, π, induction on arrival prefixes identifies the entire simulated execution with the fixed-vector execution on $\widetilde{W}, R, \pi$. Accepted labels lie outside the sacrificed mask, so their actual values are the glued coordinates. Feasibility and measurability are preserved. Even an adaptive adversary using all samples, values, and used or unused seed bits realizes one of finitely many permutations. Pointwise its reward is at least the fixed-vector minimum over those permutations. Independence and nonnegative integration therefore give

$$\begin{aligned}\mathbb{E}[\mathrm{reward}] &\geq \mathbb{E}_{\widetilde{W}} \mathbb{E}_R \min_{\pi} \mathrm{reward}_{\mathrm{fixed}}(\widetilde{W}, R, \pi) \\ &\geq C(B)\mathbb{E}[\mathrm{OPT}_M(V)] \geq C_{141}\mathbb{E}[\mathrm{OPT}_M(V)].\end{aligned}$$

The finite minimum is measurable. Tonelli's theorem justifies the iterated nonnegative integral; only the expected optimum is assumed finite, with no extra moment condition on loops. The already covered degenerate instances, ties, and zeros pass through the pointwise simulation. Together with (1), this proves Theorem 1 in the original adversarial information model. □

Appendix A. Exact scalar certificate

This is the scalar argument accompanying the accepted proof, applied to the unchanged parameters of [4]. It establishes (3)–(5) independently of numerical logarithms. With the notation of Section 1,

$$p = \frac{24383141}{67108864}, \qquad p > (3/4)^4.$$

Thus $p^{-1/4} < 4/3$. Put $A = \rho(3 + 4v/3)$. The exact rational inequalities

$$A^8 p^{105} < (av)^8, \qquad (1-p)^8 p^{113} < (bv)^8 \tag{31}$$

imply the two tilts: the first bounds $\rho p^{\mu}(p^{-1/4} + 3/v)$, and the second bounds $(1-p)p^{\nu}/v$. Taking eighth roots preserves strict inequalities because all quantities are positive. The included exact checker verifies (31) by integer and rational operations.

For $\zeta = (1-p)/(1+p) \in (0,1)$, integration of the positive geometric series gives

$$-\log p = 2\sum_{j\geq 0} \frac{\zeta^{2j+1}}{2j+1} > 2\sum_{j=0}^{3} \frac{\zeta^{2j+1}}{2j+1} > d.$$

The last comparison is rational. Direct substitution yields

$$\gamma = \frac{69959099203}{812800834560}, \qquad \varepsilon = \frac{69328715}{2294341632}, \qquad a_0 = \frac{250}{253}.$$

Exact comparison gives $0 < \gamma \leq \rho/4$ and $(1-t)(\gamma\kappa + a_0)/K < 1/4$. Also $261^{256} > 141 \cdot 256^{256}$, hence $B > 141$, and $L(\theta - 1) = 5$.

For $x > 1$, define

$$\mathcal{L}(x) = \varepsilon + Kt + \frac{Kt + K\rho/\kappa}{x-1}, \qquad s(x) = (1-h)(\gamma - \mathcal{L}(x)/h).$$

Then the attributed coefficient is

$$C(x) = \frac{(1-t)((x-1)/x)^2 t\, s(x)}{5K}.$$

The coefficient of $1/(x-1)$ in $\mathcal{L}$ is positive, so $s(B) > s(141)$. Rational arithmetic gives

$$s(141) = \frac{3932831273804099}{747774506088529920} > 0.$$

Both $((x-1)/x)^2$ and $s(x)$ are positive and strictly increasing for $x \geq 141$. Therefore

$$C(B) > C(141) = C_{141},$$

with the exact value and positive difference in (1). This proves all scalar conditions used in the fixed-vector argument. The script `checkers/check_scalar.py` checks these 15 finite arithmetic statements exactly; it does not prove a universal matroid claim.

Appendix B. The rational improvement

The original OpenAI construction gives coefficient 2^{-293} in its main argument [1]. Using the construction and quantitative refinements of Shea [3, 4], the explicit certificate here is

$$C_{141} = 165971304277649481670619/3249567587164160468118417702912 00 > 2^{-293}.$$

The adversary observes all samples, values and internal seed bits. The algorithm and full $C(B)$ formula are inherited from those sources; using $B > 141$ sharpens the displayed conservative rational certificate.

References

1. OpenAI, *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary*, OpenAI Math Release preprint, September 23, 2026, Pinned source, commit `1581d4cbdd69a0faf34b2089614fe2a814316a6a`.
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3. ———, *Sharper one-sample matroid prophet inequalities*, Repository manuscript, October 11, 2026, Pinned source, commit `8cd1395994596d89ad20c2af8b69dab8e3402ee1`.
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Mathematical supplement

The proof supplements, exact certificate data and arithmetic programs referenced by filename are included in the attached mathematical supplement (ZIP). Open the attachment using a PDF reader with file-attachment support.